Random Variables in Multiple Dimensions

Tuesday · joint, marginal, conditional, and independent

Random Vectors

- We've dealt mostly with a single random variable X that maps the sample/state space $\mathcal{S}$ into a number, X: $\mathcal{S} \rightarrow \mathbb{R}$
- An n-dimensional vector of random variables is a mapping X: $\mathcal{S} \rightarrow \mathbb{R} \times \mathbb{R} \times ... \times \mathbb{R} = \mathbb{R}^L$
- This is more complex than one dimension: **We want to track how random variables move together**
- We will look at the **joint** and **marginal** distributions, and then a different way to think of **conditional probability** and **independence** in these terms

1. Joint, Marginal, and Conditional

Joint, Marginal, Conditional Probability Distributions

- **Joint**: the probability of both
  $m_{X,Y}(x, y) = \text{pr}[X = x, Y = y]$ "P[A AND B]"
- **Marginal**: "integrate X out"
  $m_Y(y) = \sum_x m_{X,Y}(x, y)$ "P[B]"
- **Conditional**: select a row, then renormalize "P[A|B]"
  $\text{pr}[Y = y|X = x] = \frac{m_{X,Y}(x, y)}{m_X(x)} = \frac{\text{joint}}{\text{marginal}}$
- Empirically, every m is a proportion of rows in the data

	Y = 0	**Y = 1**	**Marginal**
X = a	joint	joint	row sum
X = b	joint	joint	row sum
Marginal	col sum	col sum	1

Conditional on X = a: divide row a by its row sum

You can get this table by using pd.crosstab on X and Y

$$\mathit{CONDITIONAL} = \frac{\mathit{JOINT}}{\mathit{MARGINAL}}$$

Worked Example: Calculate the joint & marginal distributions

Person	Smoker	Sick
1	Yes	Yes
2	Yes	Yes
3	Yes	No
4	No	No
5	No	Yes

X = Smoker, Y = Sick

- **Step 1: count** each (Smoker, Sick) combination. Add row and column totals

	Sick	Not sick	Total
Smoker	2	1	3
Non-smoker	1	1	2
Total	3	2	5

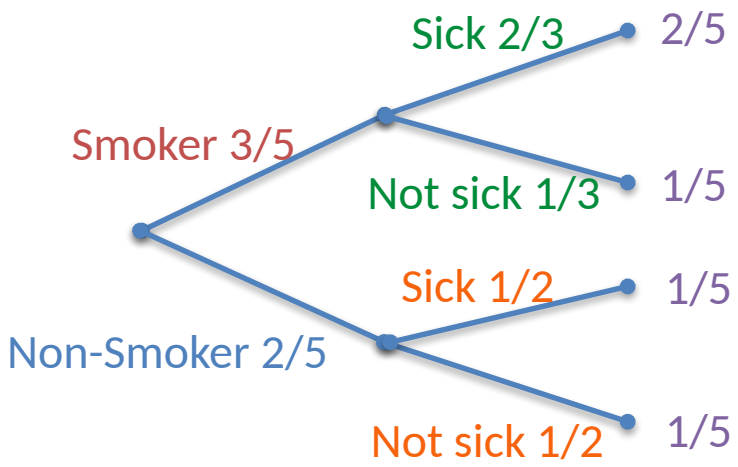
- **Step 2: joint** = count / 5. The **margins** are the row and column sums

	Sick	Not sick	Marginal
Smoker	2/5	1/5	3/5
Non-smoker	1/5	1/5	2/5
Marginal	3/5	2/5	1

The whole table sums to 1

- **Joint**: pr[Smoker, Sick] = 2/5. Two of the five people are both
- **Marginal**: pr[Sick] = 2/5 + 1/5 = 3/5. Sum down the Sick column, ignoring smoking
- In general, marginal: To describe just one of the two variables, sum over the other

Worked Example: Conditional and Independence



Person	Smoker	Sick
1	Yes	Yes
2	Yes	Yes
3	Yes	No
4	No	No
5	No	Yes

	Sick	Not sick	Marginal
Smoker	2/5	1/5	3/5
Non-smoker	1/5	1/5	2/5
Marginal	3/5	2/5	1

- **Step 3: conditional** = divide each row of the joint by its row marginal

pr[· | row]	Sick	Not sick	Row sum
Smoker	2/3	1/3	1
Non-smoker	1/2	1/2	1

- pr[Sick | Smoker] = 2/3, but pr[Sick | Non-smoker] = 1/2
- Code Sick = 1, not sick = 0: the **conditional expectation** E[Sick | Smoker] = 2/3 too

- **Step 4: independent?** If yes, every conditional row would equal the marginal pr[Sick] = 3/5
- 2/3 and 1/2 both differ from 3/5, so **not independent** in this sample
- Same check with the joint: pr[Smoker, Sick] = 2/5 = 0.40, but pr[Smoker] × pr[Sick] = 3/5 × 3/5 = 0.36
- **Direction matters**: pr[Smoker | Sick] divides the Sick **column** by 3/5: (2/5)/(3/5) = 2/3. It equals pr[Sick | Smoker] here only because both marginals are 3/5

By Hand: Rain or Shine

Ten days of weather (S = sun, R = rain):

S S S R R R S S R R

- List the 9 (today, tomorrow) pairs and fill in the **joint** table of proportions
- Compute the **marginal** of today
- Compute the **conditional** PMF of tomorrow given today
- Is tomorrow **independent** of today?

		tomorrow		
		S	**R**	**Marginal**
today	**S**	?	?	?
	R	?	?	?
	Marginal	?	?	1

rows are the current observation and **columns are the following observation.**

With code: The Rain/Sun Cross Tab

```
weather = list('SSSRRRSSRR')
today    = pd.Series(weather[:-1], name='today')
tomorrow = pd.Series(weather[1:],  name='tomorrow')

# joint, with marginals
pd.crosstab(today, tomorrow, normalize=True, margins=True)

# conditional
pd.crosstab(today, tomorrow, normalize='index')
```

- Shift the sequence by one to line up the 9 (today, tomorrow) pairs
- `normalize=True`: divide by all 9 pairs (**joint**). `margins=True` adds the **marginals**
- `normalize='index'`: divide each row by its row sum (**conditional**)

today \ tomorrow	R	S	All
R	0.3333	0.1111	0.4444
S	0.2222	0.3333	0.5556
All	0.5556	0.4444	1.0000

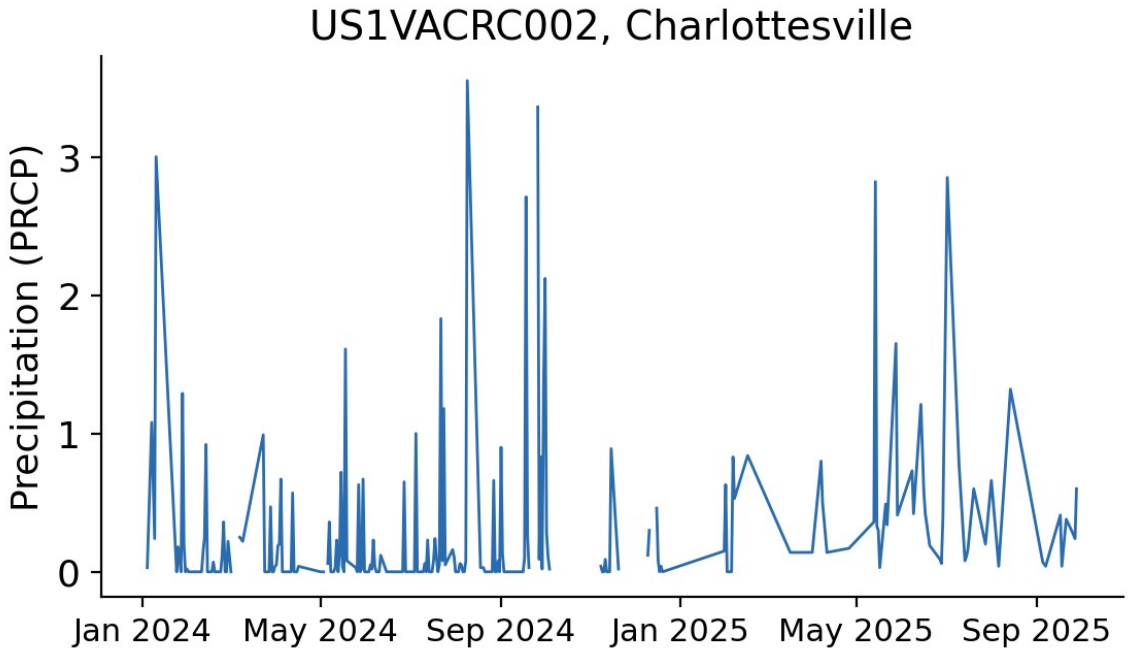
Joint, with margins

today \ tomorrow	R	S
R	0.7500	0.2500
S	0.4000	0.6000

Conditional: each **row** sums to 1. pandas sorts R before S

Real Weather: Charlottesville, Dry or Rainy?

- One station, 271 recorded observations, Jan 2024 to Sep 2025
- **Dry** if precipitation $X_t = 0$, **Rainy** if $X_t > 0$
- t + 1 is the **next recorded observation**, not necessarily the next calendar day: the station's dates have gaps
- Missing precipitation (7 rows) stays missing, not Dry




```
df['y_t']   = df['PRCP_gtr0'].map({False:'Dry', True:'Rainy'})
df['y_tp1'] = df['PRCP_gtr0'].shift(-1).map({False:'Dry', True:'Rainy'})   # shift into the future
pd.crosstab(df['y_t'], df['y_tp1'])
```

`shift(-1)` places the following row's value beside the current row's value. Same trick as `weather[1:]` in the toy example

Real Weather: Counts, Joint, and Marginals

y_t \ y_t+1	Dry	Rainy
Dry	86	35
Rainy	37	98

Raw counts: 256 complete pairs

JOINT DISTRIBUTION

y_t \ y_t+1	Dry	Rainy	All
Dry	0.336	0.137	0.473
Rainy	0.145	0.383	0.527
All	0.480	0.520	1.000

`normalize=True, margins=True`: the **whole table** sums to 1

CONDITIONAL PMF

y_t \ y_t+1	Dry	Rainy
Dry	0.711	0.289
Rainy	0.274	0.726

`normalize='index'`: each **row** sums to 1

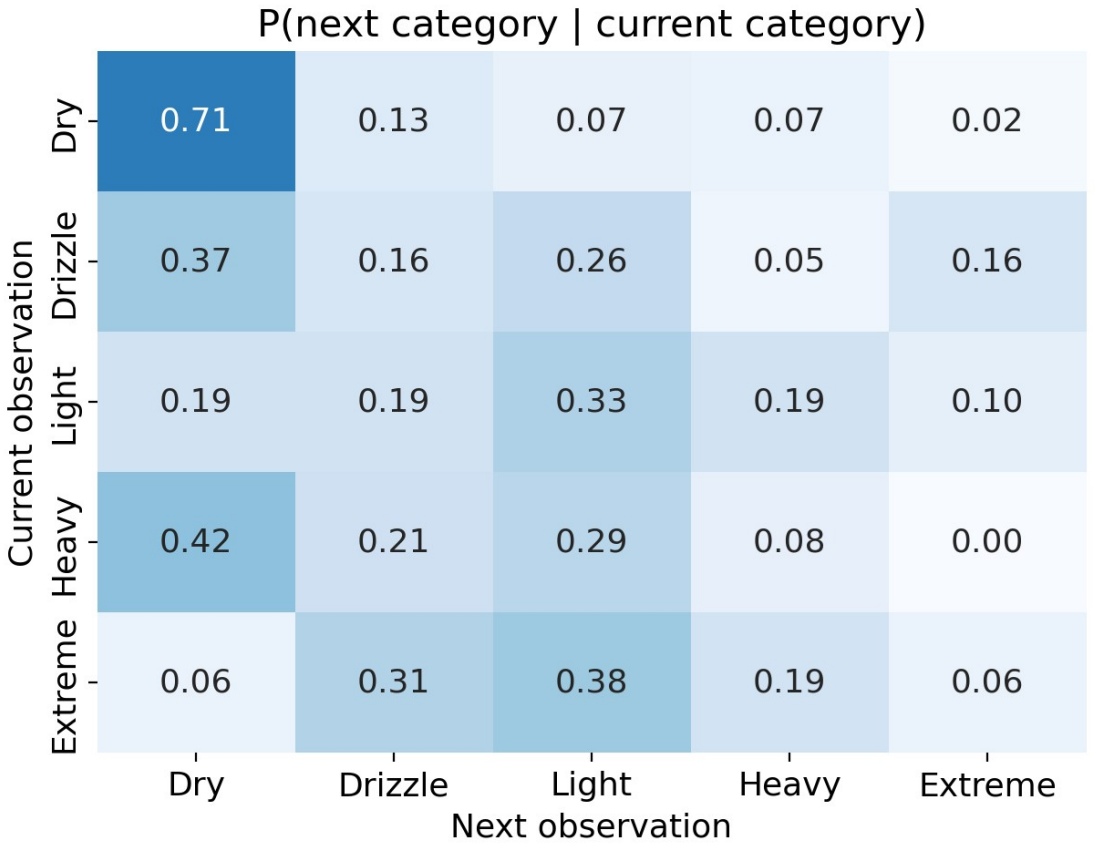
- Each cell in the joint PMF answers a question about **both** observations: 38% of pairs are rainy followed by rainy
- The conditional answers: **given** a rainy observation, 73% of the time the next one is rainy. Given dry, only 29%
- Is the next observation **independent** of the current one? Compare each conditional row with the marginal of y_{t+1}

Real Weather: Five Categories

```
df['PRCP_bin'] = pd.cut(
    df['PRCP'],
    bins=[-0.001, 0, 0.1, 0.5, 1, np.inf],
    labels=['Dry', 'Drizzle', 'Light',
            'Heavy', 'Extreme'])
df['y_t']   = df['PRCP_bin']
df['y_tp1'] = df['PRCP_bin'].shift(-1)

pd.crosstab(df['y_t'], df['y_tp1'],
            normalize='index')
```

- **Binning** keeps some information about the amount: Dry/Rainy threw it away
- Read the Dry row: **given** a dry observation, the distribution of the next category
- Rare categories (16 Extreme, 24 Heavy pairs) give noisy rows



CONDITIONAL PMF

$$\widehat{P}(Y_{t+1} = j \mid Y_t = i) = \frac{\widehat{P}(Y_t = i, Y_{t+1} = j)}{\widehat{P}(Y_t = i)} = \frac{n_{ij}}{\sum_k n_{ik}}.$$

Conditional Expectation: Tomorrow's Amount

- A conditional PMF gives probabilities of categories. A **conditional expectation** gives an **average numerical outcome** after restricting to a category:

$$E[X_{t+1} \mid Y_t = i]$$

```
df['PRCP_tp1'] = df['PRCP'].shift(-1)
df.groupby('y_t', observed=False)['PRCP_tp1'].describe()
```

- Average the **actual amounts** of the next observation, zeros included, not the bin labels
- A small mean can come with occasional large amounts: look at the count and spread too

Current y_t	count	mean	std	max
Dry	121	0.097	0.266	1.61
Drizzle	43	0.419	0.758	3.55
Light	52	0.448	0.705	3.00
Heavy	24	0.158	0.233	0.84
Extreme	16	0.389	0.519	2.12

$E[X_{t+1} \mid Y_t = i]$ is the **mean** column (precipitation units)

The **joint PMF** describes two categories together.

Summing its cells across a row/column gives a **marginal PMF**

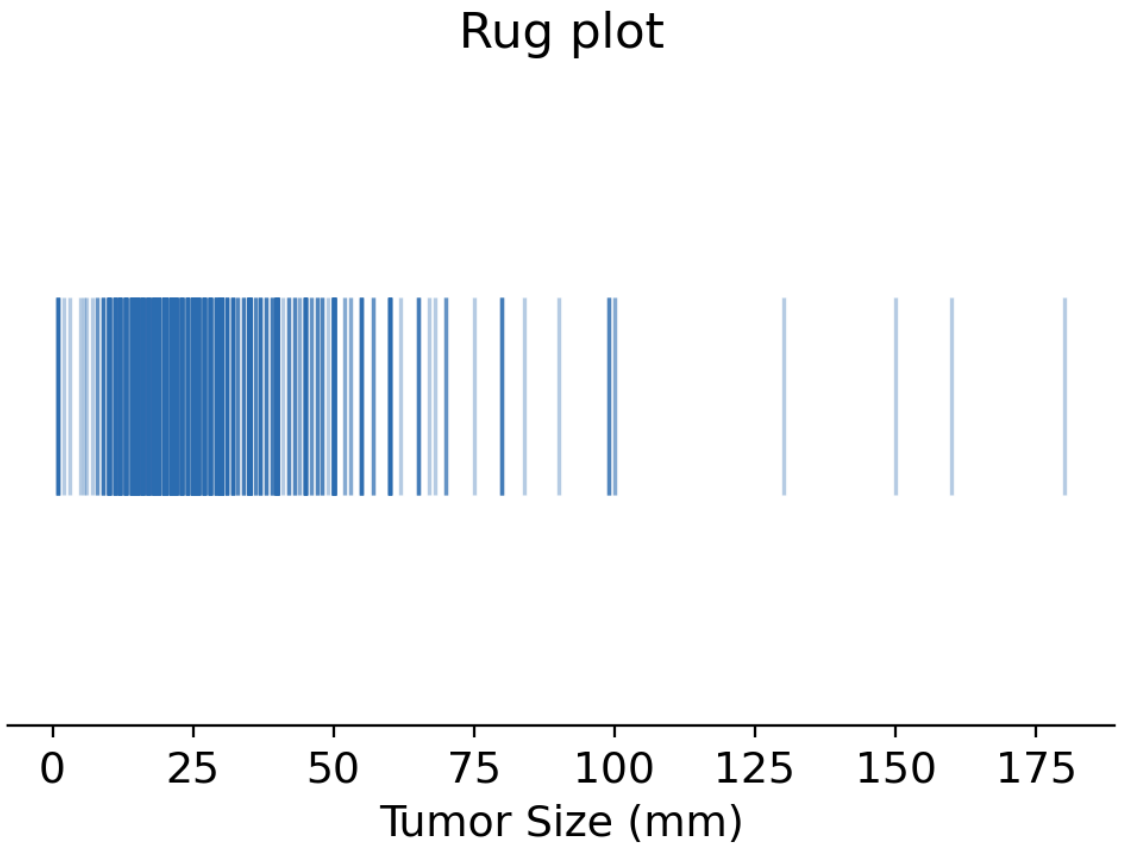
Dividing a row by its marginal probability gives a **conditional PMF**.

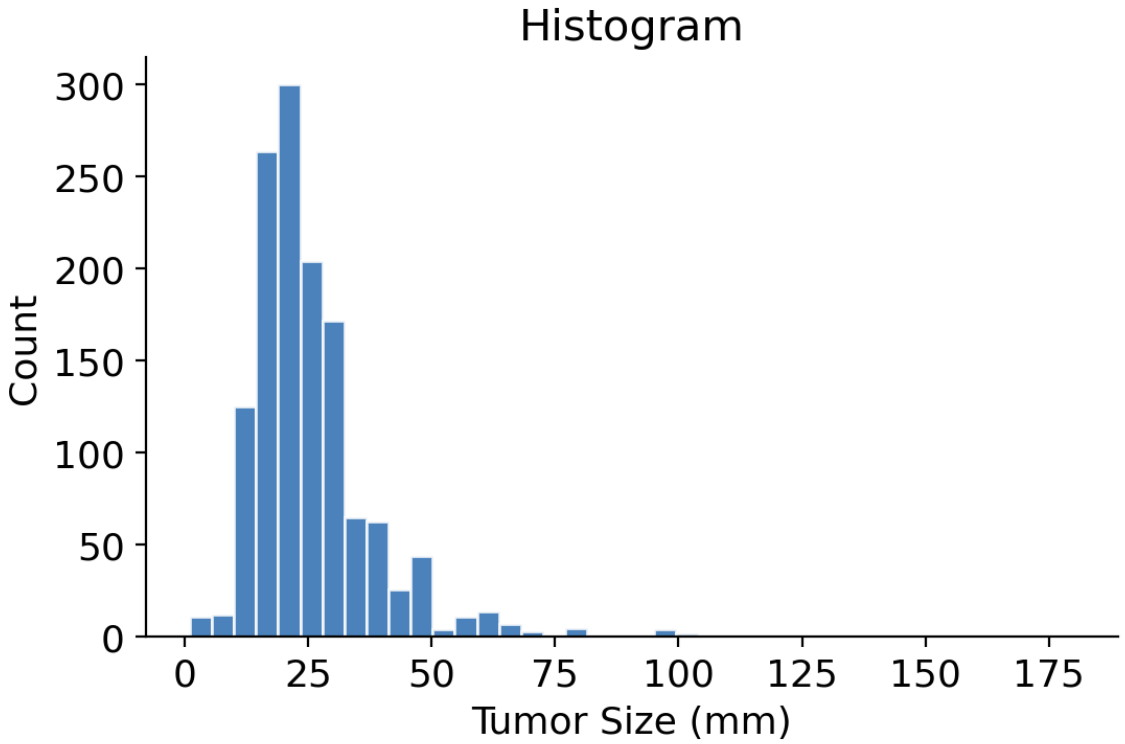
To obtain a **conditional expectation** of precipitation amount, we return to the numerical measurements and average within the selected current category.

For an indicator outcome, these ideas meet directly: the conditional expectation of "the following observation is rainy" equals the conditional probability that it is rainy. For precipitation amount, the conditional expectation is an average amount, including zeros.

2. Densities and Independence

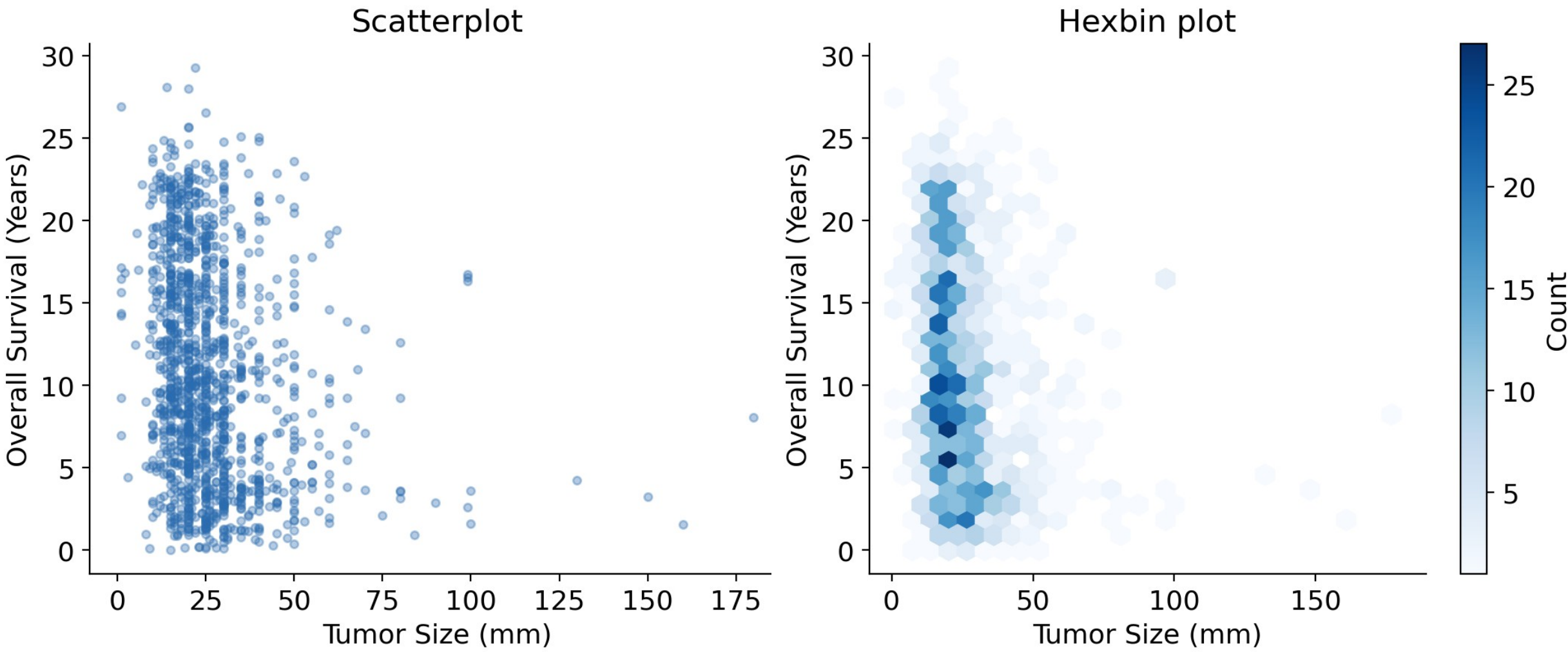
Which Do You Like Better?





Same 1,343 tumor sizes. Where are most of the observations?

Which Do You Like Better?



Tumor size vs. survival. A scatterplot is a two-dimensional rug plot: overlapping points hide where the mass is

Independence

- Shrink the hexes to points and the counts become a **joint density**
- (X_1, ..., X_n) are **independent random variables** if the joint density decomposes into a product of marginals:

$$f_{X_1,X_2,...,X_n}(x_1, x_2, ..., x_n) = f_{X_1}(x_1) \times f_{X_2}(x_2) \times ... \times f_{X_n}(x_n)$$

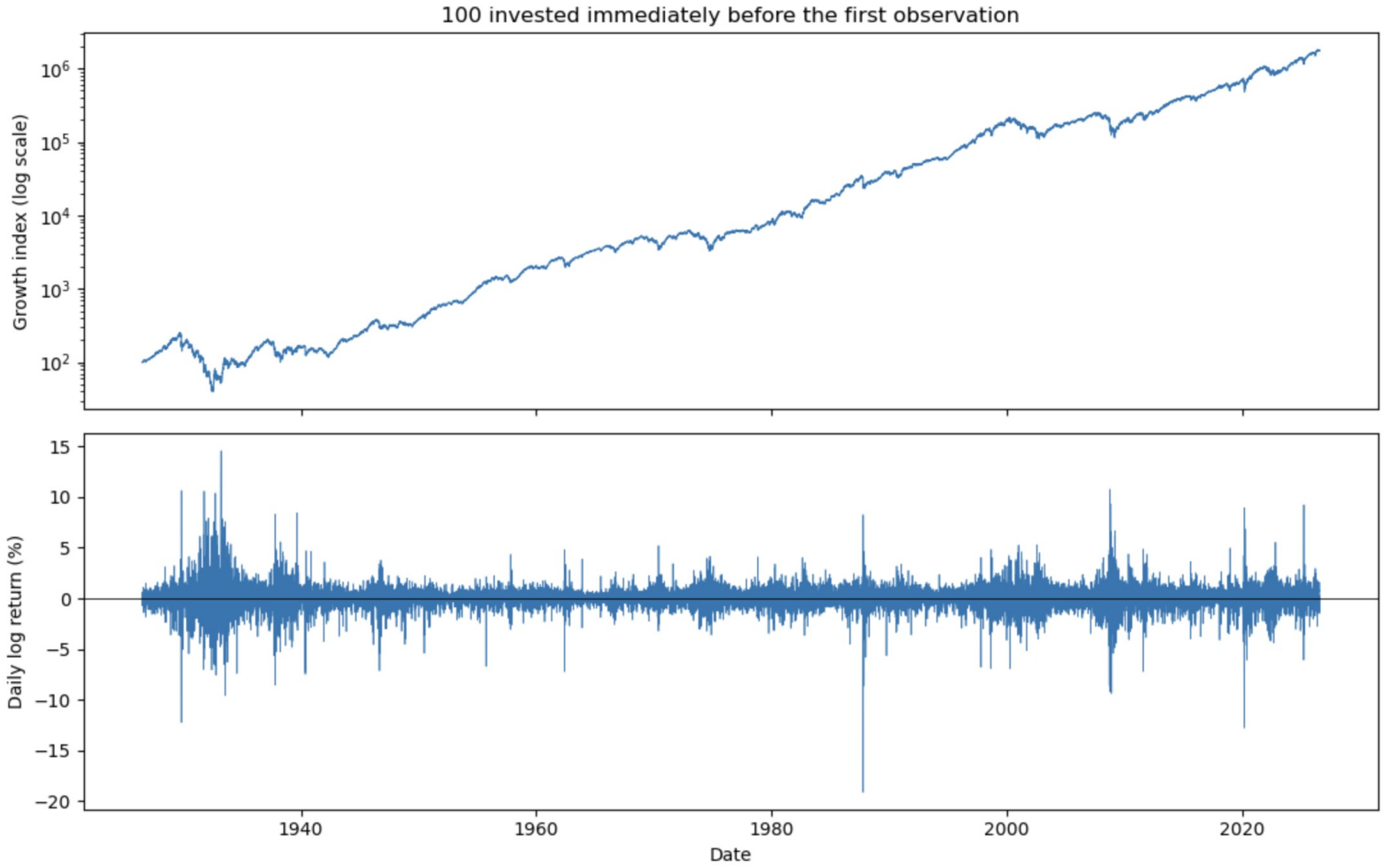
- For tables, the same idea:

$$\text{pr}[Y = y | X = x] = \text{pr}[Y = y] \quad \text{for all } x, y$$

- If independent, **every conditional row equals the marginal**. Knowing X tells you nothing about Y

Market Returns: Up or Down Tomorrow?

- 26,296 daily US market returns, 1926 to 2026. Use **log returns**: $r_t = \log(1 + R_t)$
- Pair each trading day with the **next trading day**. Same two-category question as Dry/Wet



direction_tp1 / direction_t	Down / flat	Up
Down / flat	5909	5999
Up	5999	8388

Market Returns: Up or Down Tomorrow?

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- Pair each trading day with the **next trading day**. Same two-category question as Dry/Wet

Today \ Next	Down / flat	Up	All
Down / flat	0.225	0.228	0.453
Up	0.228	0.319	0.547
All	0.453	0.547	1.000

Joint with margins. The **whole table** sums to 1

Today \ Next	Down / flat	Up
Down / flat	0.496	0.504
Up	0.417	0.583

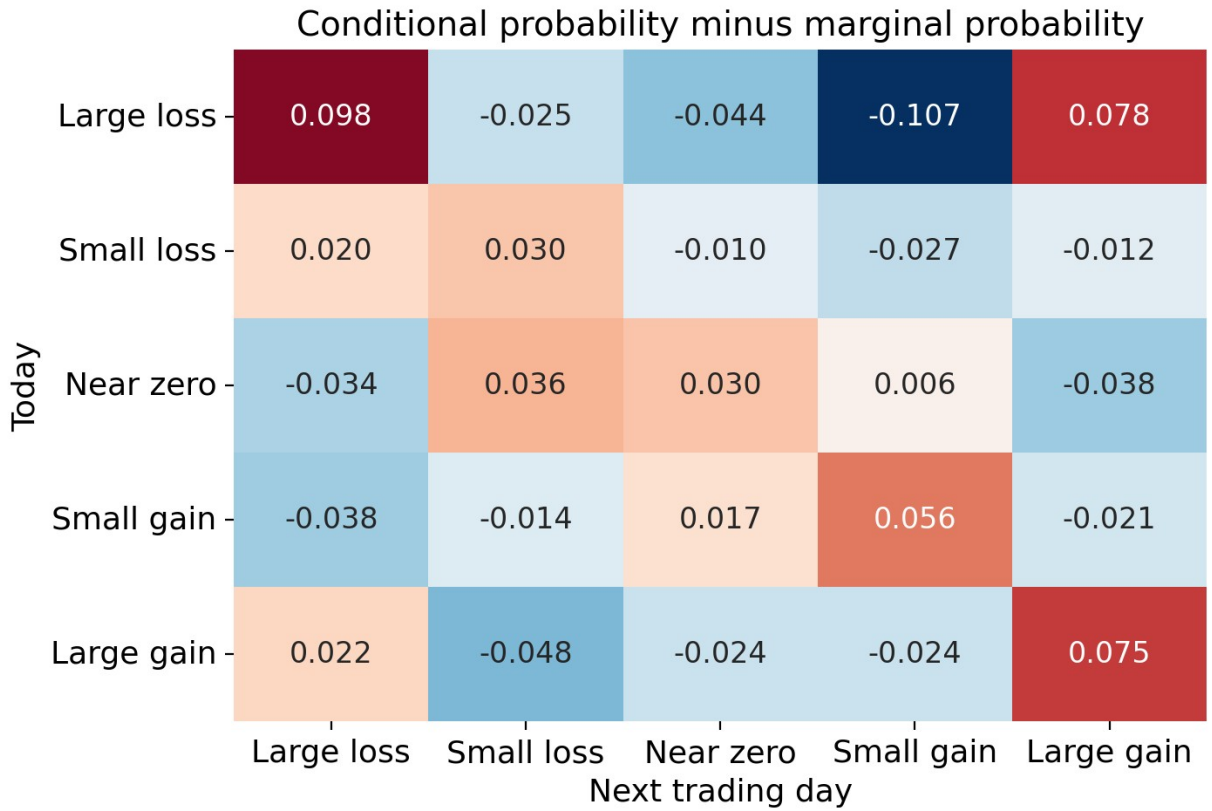
Conditional. Each **row** sums to 1

- Marginal: the market is up on 54.7% of days. Given today is up, tomorrow is up 58.3% of the time; given down, 50.4%
- The rows are close to the marginal but not equal. Is that a big difference?

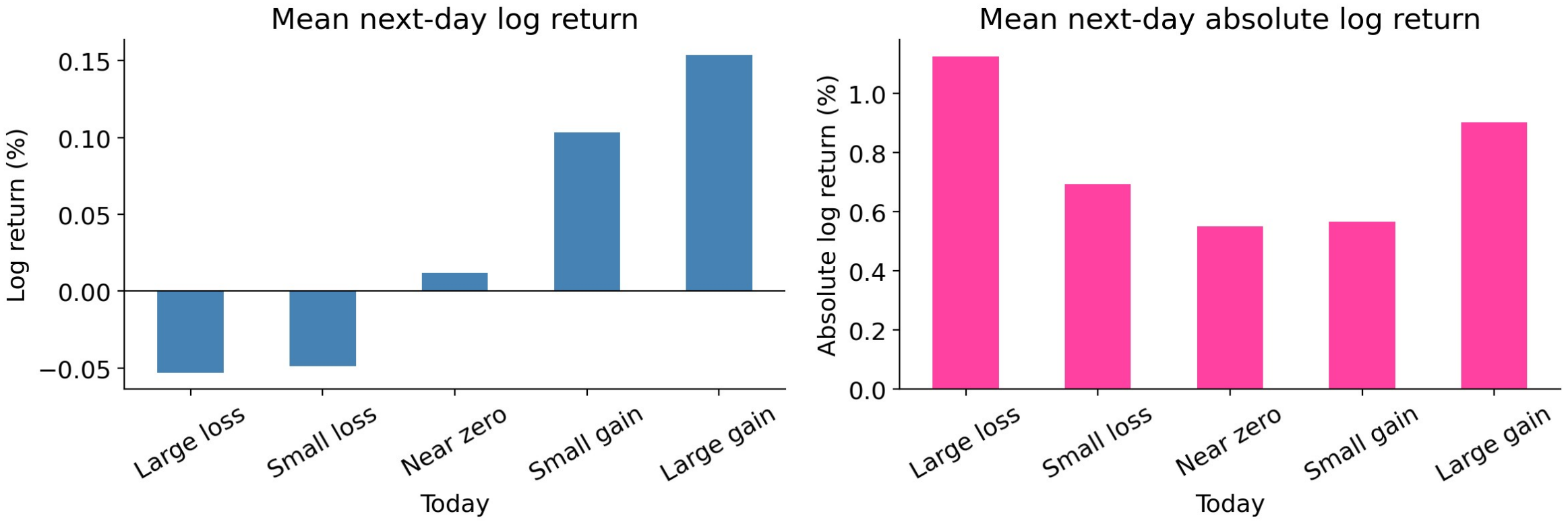
Market Returns: Five Categories vs. Independence

- Bin log returns (%): **Large loss** ≤ –1 < **Small loss** ≤ –0.1 < **Near zero** ≤ 0.1 < **Small gain** ≤ 1 < **Large gain**
- Compute the conditional PMF of tomorrow's bin given today's, then **subtract the marginal** of tomorrow's bin
- If today and tomorrow were independent, every cell would be **0**
- Red: more likely than the marginal. Blue: less likely. Units are probability points (e.g. table is pr[tomorrow's bin | today's bin] − pr[tomorrow's bin] so a large loss tomorrow is **9.8 percentage points** more likely after a large loss today)

```
return_conditional.sub(next_category_pmf,
                       axis='columns')
```



Conditional Expectation: Volatility Clustering



- $E[r_{t+1} \mid \text{today's bin}]$ (left) is close to 0 for every bin: direction is hard to predict
- $E[|r_{t+1}| \mid \text{today's bin}]$ (right): after a **large loss**, the next day's average move is 1.13%, about double the 0.55% after a near-zero day. Big moves follow big moves

Points to Keep in Mind

- **Joint and conditional probabilities use different denominators.** The joint sums to 1 over the whole table. Each row of the conditional sums to 1 on its own
- **Conditional means retain the numerical outcome.** Category probabilities and average amounts answer different questions
- **Binning is a choice.** Different thresholds change the summaries. Rare categories give less precise estimates
- **These are descriptive summaries.** Differences between conditional rows describe association in this sample, not a causal effect